You check your phone and see a mobile money transfer of `$25`, a weather app showing `32°C`, and a message from school about your attendance. Each of those is a piece of data, a small fact about something that happened. Before you analyze data, build dashboards, or choose a career in data, you need a clear answer to a simple question: **what is data?**

# What Is Data?

Lesson 1 of 5

BY THE END

**You can explain what data is, and tell data, information, and insight apart.**

# Data: raw facts about the world

**Data** is a collection of facts, measurements, or observations about the world. On its own, data usually does not tell a full story. It is the raw material we use to understand what is happening.

# Examples of data

- `$25` - a mobile money transfer amount
- `$2.50 per kg` - the market price of rice
- `present` - a student's attendance record
- `$49.99` - a product price
- `32°C` - a temperature reading

Each example is a single fact. Collect enough of them and patterns start to appear.

# Data vs information vs insight

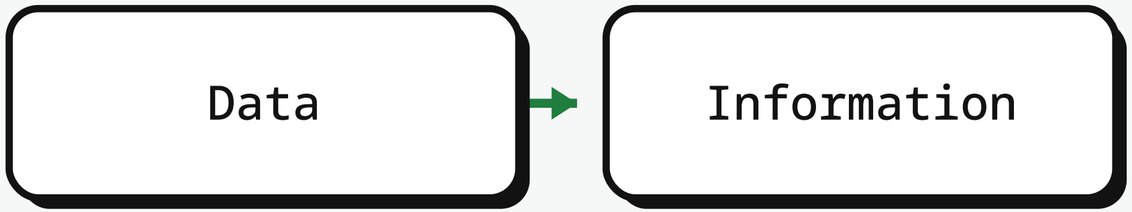
These three terms are related, but they are not the same.

| TERM | MEANING | EXAMPLE |
| --- | --- | --- |
| Data | Raw facts | Sales were `$1,200`, `$980` and `$1,450` on three days |
| Information | Data organized to answer a basic question | Sales increased from Monday to Wednesday |
| Insight | A useful conclusion that can guide action | Promotions on Wednesday may be driving higher sales |

**PLATE I - FROM FACTS TO A DECISION**

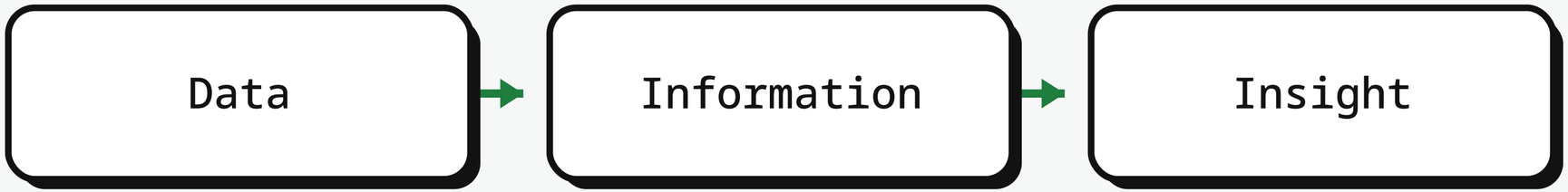
Raw facts

**PLATE I - FROM FACTS TO A DECISION**




Data organized to answer a basic question

**PLATE I - FROM FACTS TO A DECISION**




Each step throws information away and adds meaning. You cannot skip one: an insight with no data underneath it is just an opinion.

## A useful conclusion that can guide action

# Think of it like cooking

| DATA | INFORMATION | INSIGHT |
| --- | --- | --- |
| The ingredients, before anyone has decided what to make. | A prepared meal you can actually understand. | Knowing what to cook next time, based on what people liked. |

# Where you see data every day

Data is not only found in spreadsheets or databases. You create and use data constantly:

- Mobile money transactions and bank transfers
- School grades and attendance records
- Market prices at a local shop
- Likes and comments on social media
- Steps tracked by a fitness app
- Weather forecasts

Businesses, schools, hospitals, and governments all rely on data to make decisions.

KEY TERMS

| TERM | DEFINITION |
| --- | --- |
| Data | Raw facts, measurements, or observations |
| Information | Data that has been organized to convey meaning |
| Insight | A meaningful conclusion, supported by data, that helps guide a decision |